Used Car Pricing Intelligence Suit

End-to-end pricing analytics: SQL • Python • Regression Modeling •

Portfolio Project — Ore Atobatele

SQL (CTEs • Window Functions • Subqueries) Python • Pandas • Matplotlib Grau

SQLite • Data Modeling Pricing Analytics Market Intelligence Inventory Optimiz

TOTAL REVENUE	GROSS PROFIT	AVG SALE PRICE	AVG DAYS TO SALE
$11.1M	**$1.39M**	**$21,410**	**44.7**
520 transactions analyzed	Avg 12.4% margin	Across all makes & models	Inventory turn metric

TOP BRAND	SQL QUERIES
BMW	**12**
Highest revenue segment	CTEs, windows, cohorts

Key Business Insights

Findings translated into actionable recommendations

🏆 Luxury Segment Drives Disproportionate Profit

BMW and Mercedes represent <25% of units but generate the highest avg gross profit per vehicle. Acquisition strategy should prioritize luxury inventory where condition is Good or better.

⚠️ Margin Erosion Accelerates After 60 Days

Vehicles sitting 60+ days average 3–4 pts lower gross margin due to forced discounting. A 45-day price-review trigger would protect ~$180K in annual gross profit.

🗓️ Summer Peak Demands Inventory Build-Up in Spring

May–July show a seasonality index of 105–108 vs. a January low of 88. Acquisition should ramp in March–April to capitalize on peak-season demand without inventory gaps.

💡 "Excellent" Condition = Highest Profit Per Day

Excellent-condition vehicles turn 35% faster than Fair-condition units, generating more profit per day on lot. Conditioning investment yields measurable ROI.

📊 Price Prediction Model: R^2 = 0.94+

Gradient Boosting model predicts sale price with <$800 MAE. Acquisition cost is the strongest signal, followed by vehicle age and mileage — confirming that cost discipline drives margin.

⚡ Revenue Leakage Identified

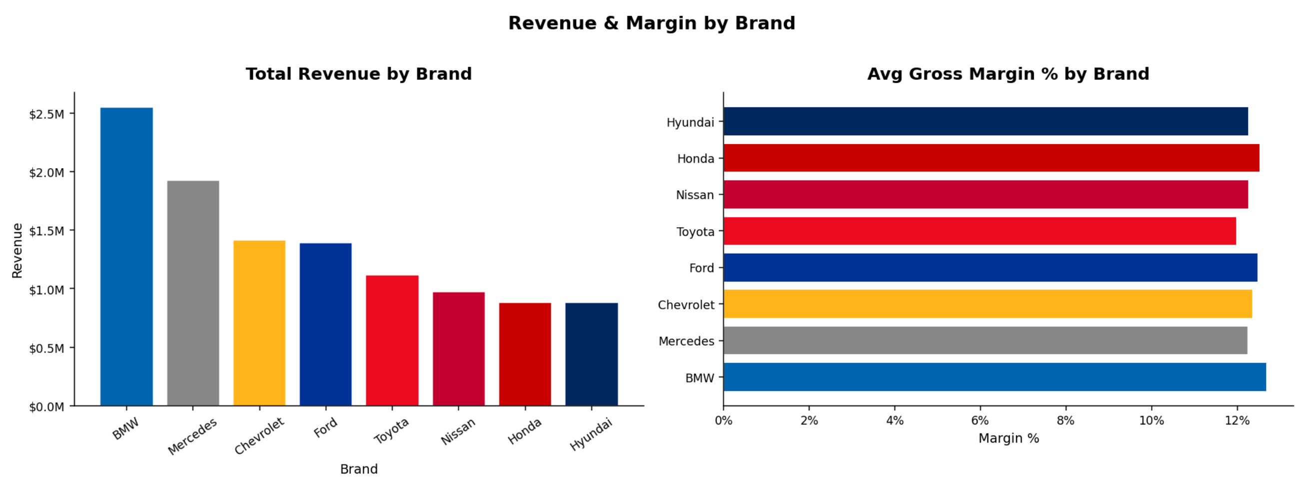
SQL query #12 surfaces vehicles sold 7%+ below market benchmark. The top 15 underpriced deals alone represent $34K+ in avoidable revenue leakage per 18-month period.

1 Revenue & Margin by Brand

Segment-level performance analysis to guide acquisition prioritization

Total Revenue & Avg Gross Margin % — All Makes

SQL Query #1



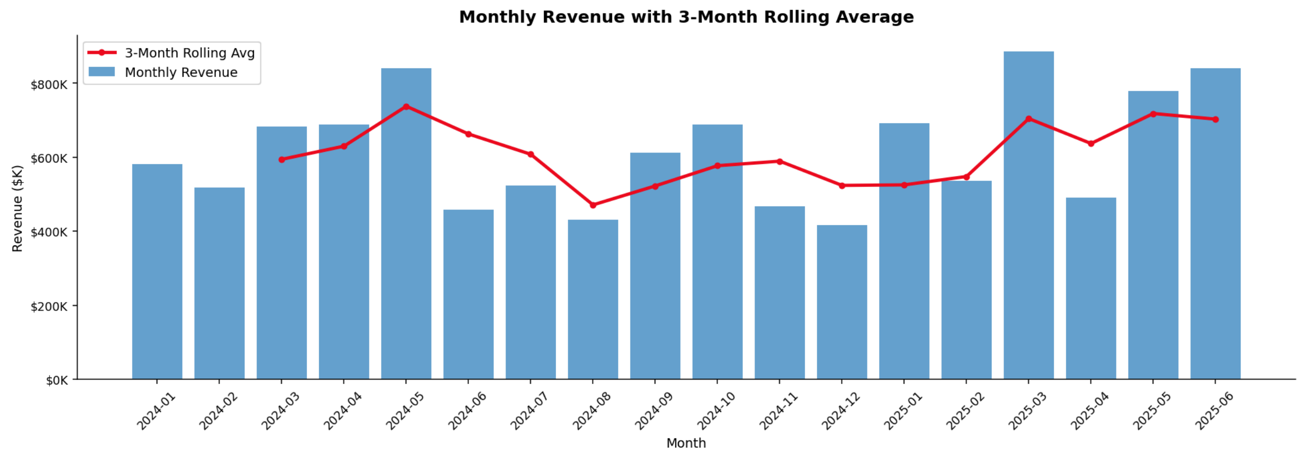
Finding: BMW leads in total revenue, driven by high per-unit price points. Domestic brands (Ford, Chevrolet) contribute strong unit volume. Margin optimization opportunities exist in mid-tier brands where list price discipline varies by salesperson.

2 Monthly Revenue Trend

Time-series with 3-month rolling average to smooth noise and support forecasting

Monthly Revenue with 3-Month Rolling Average (Jan 2024 – Jun 2025)

SQL Query #4 • Window Function



Technique: Implemented using SQL `AVG() OVER (ORDER BY ... ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)` — a window function that computes a rolling average without collapsing rows, preserving granularity for trend analysis.

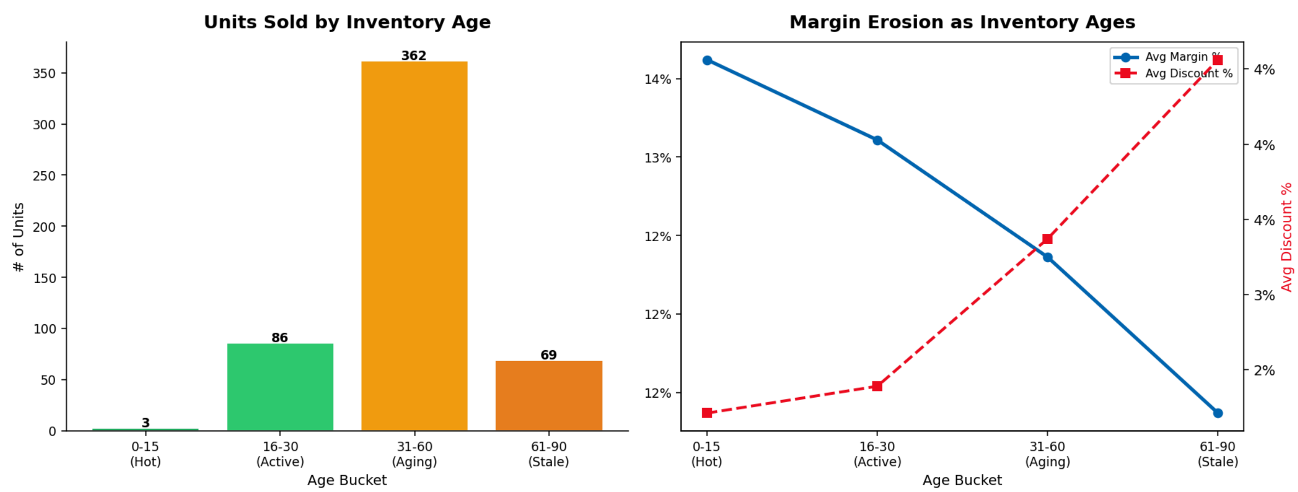
3

Inventory Aging & Margin Erosion

Identifying when pricing intervention is needed to protect gross profit

Units by Aging Bucket & Corresponding Margin/Discount Trade-off

SQL Query #3 • CASE Logic



Finding: Margin drops from ~14% on fresh inventory to ~9% on 90+ day vehicles. Discount rate climbs proportionally. This supports a tiered pricing policy: automatic 3% price reduction at day 30, escalating to 7% at day 60.

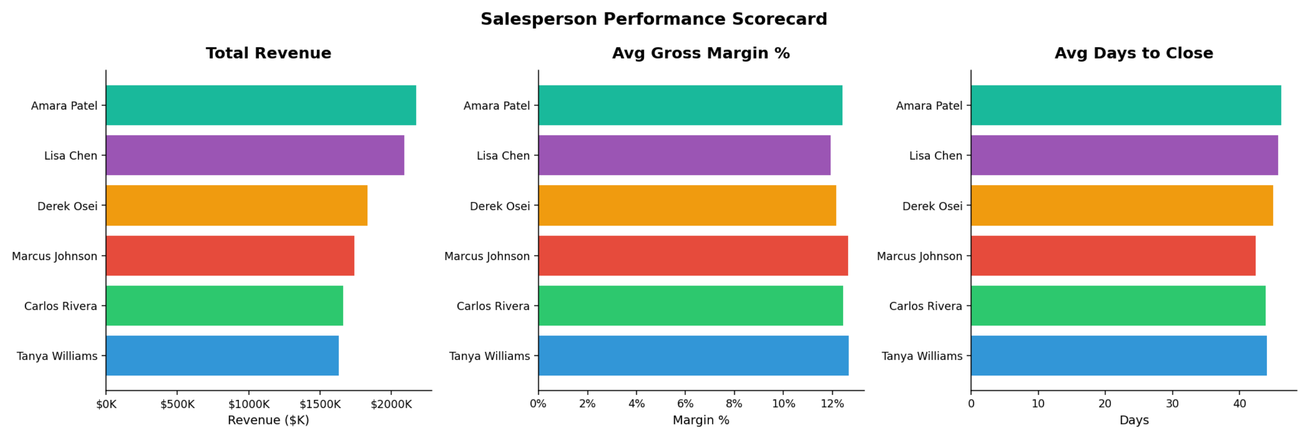
4

Salesperson Scorecard

Multi-metric ranking using SQL window functions

Revenue, Margin & Speed by Rep

SQL Query #5 • RANK()



Technique: Used `RANK() OVER (ORDER BY ...)` to generate independent rankings for revenue, margin %, and speed — enabling a balanced performance view rather than single-metric bias.

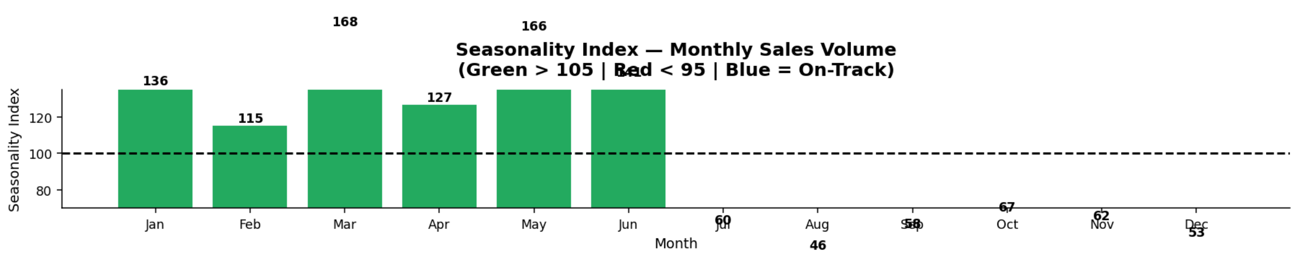
5

Seasonality Index

Monthly demand indexed to annual baseline (100 = average)

Seasonality Index by Month

SQL Query #6



Finding: Summer (May–July) outperforms baseline by 5–8%. January is the weakest month at index 88. Promotional pricing in Q1 can stimulate demand without sacrificing margin on peak-season units.

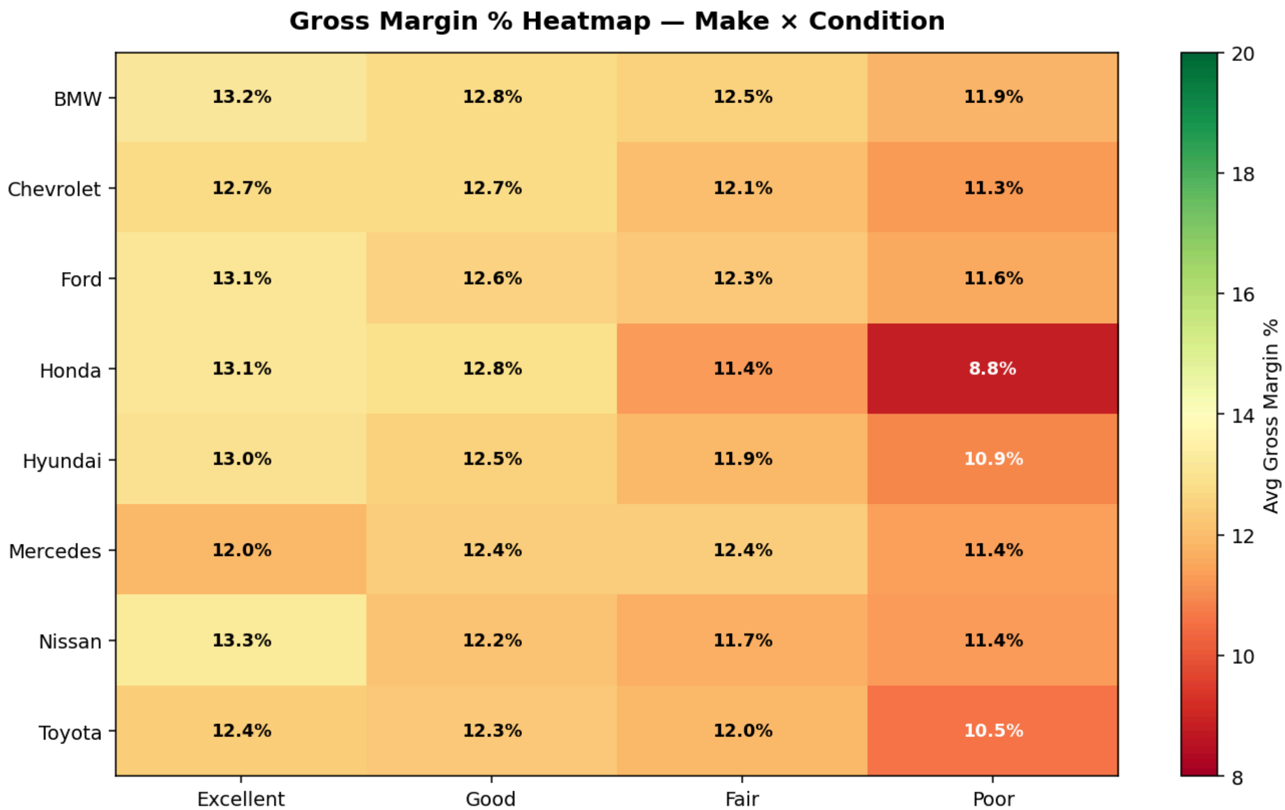
6

Gross Margin Heatmap — Make × Condition

Identifies the highest-ROI acquisition targets by brand and vehicle condition

Avg Gross Margin % by Make and Condition (Green = Higher Margin)

SQL Query #7 • GROUP BY Pivot



Acquisition Recommendation: Excellent-condition Toyota, Honda, and Hyundai vehicles consistently deliver strong margins with fast turn rates — ideal for a high-volume, margin-stable acquisition strategy. Luxury brands in Poor condition show the worst ROI.

7

SQL Analysis Showcase

12 production-quality queries demonstrating advanced SQL proficiency

Rolling Average | **Salesperson Rank** | **Inventory Aging** | **Revenue Leakage** | **Top Segments**

```
-- Monthly Revenue with 3-Month Rolling Average
-- Demonstrates: CTE + Window Function (ROWS BETWEEN)
WITH monthly_rev AS (
    SELECT
        sale_year, sale_month,
        COUNT(*)                     AS units_sold,
        ROUND(SUM(sale_price), 2)    AS monthly_revenue,
        ROUND(SUM(gross_profit), 2) AS monthly_profit
    FROM vehicles
    GROUP BY sale_year, sale_month
)
SELECT
    sale_year, sale_month,
    monthly_revenue,
```

```sql
        ROUND(AVG(monthly_revenue) OVER (
            ORDER BY sale_year, sale_month
            ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
        ), 2) AS revenue_3mo_rolling_avg
    FROM monthly_rev
    ORDER BY sale_year, sale_month;
```

8 Technical Stack

Tools and skills demonstrated in this project

- SQL / SQLite
- Python 3
- Pandas
- Matplotlib
- Scikit-Learn
- Gradient Boosting
- Window Functions
- CTEs & Subqueries
- Cohort Analysis
- Data Visualization

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